Survival-supervised deconvolution identifies a reproducible tumor–stroma prognostic axis in pancreatic cancer

Amber M. Young[a], Alisa Yurovsky[b], Xianlu Laura Peng[c,d], Jen Jen Yeh[c,d,e], Didong Li[a], Naim U. Rashid[a,d,*]

[a]Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[b]Department of Biomedical Informatics, Stony Brook University, Stony Brook, NY 11794, USA

[c]Department of Pharmacology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[d]Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[e]Department of Surgery, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[*]Corresponding author: nur2@email.unc.edu

Abstract

Clinically relevant transcriptomic programs are usually established by unsupervised discovery, post hoc outcome testing and repeated cross-study validation. We developed DeSurv, a survival-supervised matrix factorization that makes clinical relevance part of gene-program discovery while keeping the learned gene weights fixed for testing in new cohorts. In pooled TCGA and CPTAC cohorts, DeSurv recovered a three-program tumor–stroma architecture across patients, in which a dominant survival-aligned direction linked classical tumor identity with tumor-restraining (restCAF) stroma, alongside separate tumor-promoting (proCAF) stromal and basal-like tumor programs. The frozen predictor generalized across five external validation datasets spanning four independent patient cohorts (HR per SD, 1.42; 95% CI, 1.26–1.59) and remained prognostic after PurIST and DeCAF adjustment. Higher-rank NMF and conventional supervised models achieved comparable discrimination but not the same compact biological decomposition. DeSurv therefore makes clinical relevance a criterion that shapes discovery rather than one applied afterwards, separating what a method predicts from how it represents the biology.

Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal solid tumors, with few durable responses to existing therapies[1]. Bulk transcriptomic studies have identified clinically relevant malignant and stromal states in PDAC, including basal-like and classical tumor programs[2–4] and tumor-promoting proCAF and tumor-restraining restCAF programs[5]. Single-cell and spatial profiling have further shown that this heterogeneity extends across malignant, stromal, immune, and other microenvironmental compartments within individual tumors[6–8]. Whether these tumor and stromal programs together provide prognostic information beyond existing subtype classifications remains unclear.

Although single-cell and spatial profiling resolve cellular states in detail, large clinically annotated PDAC cohorts still rely primarily on bulk transcriptomic profiling[9,10]. Bulk profiles provide the scale needed for survival analysis but mix signals from malignant and microenvironmental cells[11];

nonnegative matrix factorization (NMF) and related deconvolution methods can separate these composite profiles into interpretable gene programs[3,12–14]. In PDAC, such methods have yielded molecular subtypes[2,4,15], virtual microdissection of basal-like and classical malignant programs[3], and compartment-specific tumor–stroma deconvolution[5,14]. The practical challenge is therefore not simply to resolve interpretable gene programs, but to identify during discovery which programs are clinically relevant and to test those same programs unchanged across independent studies.

These deconvolution approaches typically discover gene programs without using clinical outcomes and assess their prognostic relevance only afterward[2,15]. Standard NMF learns gene programs by minimizing reconstruction error, favoring patterns that capture prominent expression variation whether or not that variation is related to survival[9,16]. This distinction is particularly important in PDAC, where bulk tumors contain substantial stromal, exocrine, and other non-malignant signals that can dominate transcriptomic variation and obscure less prominent but prognostically relevant programs[4,5,17,18]. Reconstruction alone also does not uniquely determine the recovered gene programs without additional constraints[19–21], which may contribute to cross-study variability among unsupervised PDAC subtype taxonomies[22]. This non-identifiability applies to any factorization of this kind, including ours, and we address it explicitly rather than assume it away. Consequently, determining which unsupervised programs are both clinically relevant and reproducible has required extensive multi-study refinement. In PDAC, the basal-like/classical dichotomy was established and progressively validated over roughly a decade through iterative subtyping, virtual microdissection, independent bulk cohorts, single-cell analyses, and prospective clinical profiling[2–4,8,10]; comparable work is now establishing the clinical relevance of stromal CAF subtypes[5].

Conversely, survival-supervised models can identify outcome-associated expression patterns, but they often operate on thousands of highly correlated genes, making individual-gene effects difficult to estimate and interpret. Such models may predict survival without organizing the prognostic signal into biologically interpretable tumor and microenvironmental programs. Together, these limitations motivate an approach that incorporates survival information as gene programs are learned, while preserving their biological interpretability.

Here, we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox regression so that survival helps shape gene programs as they are learned. By incorporating survival during factorization, DeSurv is designed to recover survival-aligned biological structure that may be obscured when other sources of expression variation dominate. Once learned, the gene weights defining each program are held fixed, allowing the same programs to be evaluated directly in independent cohorts without refitting the factorization. DeSurv therefore changes clinical relevance from a post hoc criterion used to select among unsupervised factors into a criterion that shapes the factors themselves, while making cross-study transportability a direct test of the same frozen gene programs. We evaluate DeSurv in PDAC across five external validation datasets spanning four independent patient cohorts and use simulations with known latent structure to define when survival supervision improves recovery of prognostic programs.

In PDAC, DeSurv identifies a three-program tumor–stroma architecture that reflects coordinated cross-patient variation rather than spatial or cellular organization within individual tumors. Evaluating it alongside higher-rank NMF and conventional survival-supervised models separates what a method predicts from how it represents the underlying biology, and it is that distinction, rather than predictive superiority, that this work sets out to establish.

Results

DeSurv makes clinical relevance part of gene-program discovery

DeSurv learns gene programs by balancing reconstruction of the tumor transcriptome with association to survival. Unlike the conventional sequence of unsupervised factorization followed by post hoc clinical screening, survival directly shapes the gene programs during discovery. After training, the gene weights are fixed, allowing the same programs to be scored and tested in new tumors without refitting the factorization (Fig. 1A; Methods). We trained DeSurv in pooled TCGA and CPTAC PDAC cohorts[23,24] ($n = 273$, 139 events). The Bayesian-optimization response surface showed a broad region of high cross-validated performance at intermediate supervision and moderate rank, declining toward the unsupervised $\alpha = 0$ boundary (Supplementary Fig. S1). Cross-validated concordance changed little across the ranks examined, and Bayesian optimization with the one-standard-error rule selected a parsimonious three-program model ($k = 3$, $\alpha = 0.334$; peak cross-validated C-index, 0.655; Fig. 1B and Supplementary Fig. S5). By contrast, standard unsupervised NMF rank criteria (reconstruction residuals, cophenetic correlation and silhouette width[13,25]) gave conflicting choices in the same data (Supplementary Fig. S6), underscoring rank selection as an additional source of ambiguity in unsupervised decompositions. All model and hyperparameter selection was completed within the training cohorts before the programs were applied to external validation datasets.

Survival supervision resolves a distinct tumor–stroma program structure

To determine how survival supervision altered the biological representation, we compared DeSurv with standard NMF at the same rank ($k = 3$), thereby holding model complexity constant (a sensitivity analysis using NMF's independently selected ranks, elbow $k = 5$ and BO $k = 7$, is in Supplementary Table S5). DeSurv resolved three distinct programs (D1–D3): D1, enriched for classical tumor and restCAF-like stromal signatures; D2, enriched for proCAF-like stroma; and D3, enriched for basal-like tumor programs (Fig. 2a). Together, these programs define a tumor–stroma prognostic architecture spanning classical/restCAF-like and basal-like/proCAF-like states across patients. The classical identity of D1 was independently supported in the COMPASS cohort, where D1 scores correlated with GATA6 RNA in situ hybridization (Spearman $\rho = 0.68$, $P < 0.001$, $n = 33$; Fig. 2e).

Standard NMF produced a different organization at the same rank (Fig. 2b). Its malignant factor (N1) combined classical and basal-like tumor programs, its microenvironmental factor (N3) combined multiple stromal and immune programs[26,27], and a third factor (N2) captured exocrine-associated variation. DeSurv instead separated basal-like tumor (D3) and proCAF-like stroma (D2) while recovering the distinct classical/restCAF-like survival-aligned D1 program.

This difference reflected a selective reorganization rather than wholesale loss of expression structure. Of the three programs, D1 contributed the least to reconstructing the expression matrix, 18.8% of the total as a normalized Shapley share (Fig. 2c), yet it carried essentially all of the survival signal. Removing D1 cost 66 units of Cox partial log-likelihood, against 1.0 for D3 and 0.02 for D2, even though D3 alone accounted for 56% of reconstruction. The programs that best reconstruct the transcriptome need not be those most relevant to survival[18]. Pairwise gene-loading correlations across all 1,970 genes showed that the principal NMF tumor and microenvironmental factors were largely retained as D3 (Spearman $\rho = 0.84$) and D2 ($\rho = 0.87$), whereas D1 correlated only weakly with every matched-rank NMF factor ($\rho \leq 0.36$; Fig. 2d). Consistently, D1 was poorly reconstructable from the full set of NMF factors ($R^2 = 0.09$; Supplementary Table S2), while reconstructing the 1,970-gene analysis matrix somewhat less completely than unsupervised NMF (R^2 0.75 versus 0.81, a difference of 6.0 percentage points; Supplementary Table S3). Thus, survival supervision preserved most of the variance-dominant tumor and stromal structure, at a measurable cost in reconstruction, while reorganizing the representation to expose a distinct survival-aligned program.

Frozen survival-aligned programs retain prognostic relevance across independent PDAC cohorts

We next tested whether the programs discovered in TCGA and CPTAC retained their prognostic relevance when applied unchanged, without factor re-estimation or outcome-based selection, to five external validation datasets spanning four independent patient cohorts (Dijk, Moffitt, PACA-AU [array and RNA-seq], and Puleo; non-metastatic, treatment-naive; Supplementary Table S1), scoring each tumor by projection onto the trained programs, restricted to the top-ranked genes of each ($Z = \widetilde{W}^\top X_{\text{new}}$; Methods). Because the PACA-AU array and RNA-seq datasets profile a partially overlapping set of patients, combined analyses represented each patient once (retaining the RNA-seq profile and excluding the corresponding array measurement, consistent with the DeCAF preprocessing rule[5]), giving a combined validation population of 570 unique patients (388 deaths). The survival-aligned D1 program was protective in every validation dataset and was the only individual program associated with survival in the combined analysis. D3 was higher-risk in four of the five datasets but did not reach significance when combined, and the proCAF-like D2 program was near-null and more variable (Fig. 3a). The frozen multivariable DeSurv linear predictor (oriented so that higher values indicate higher risk) was associated with overall survival in a cohort-stratified analysis (pooled HR per SD = 1.42, 95% CI 1.26–1.59, $P = 5.4 \times 10^{-9}$; Table 1). At the same three-program complexity, the corresponding standard NMF predictor showed a substantially weaker association (HR per SD = 1.07, 95% CI 0.96–1.19, $P = 0.22$). These analyses therefore evaluate the same programs discovered in training rather than rediscovering, selecting or matching factors separately within each validation dataset.

We then asked whether comparable prognostic performance could be achieved without survival-supervised deconvolution. Increasing the rank of standard NMF substantially improved external discrimination: NMF at $k = 7$ matched or exceeded DeSurv's concordance in most validation cohorts and, like DeSurv, remained prognostic after simultaneous adjustment for PurIST and DeCAF (Table 1; Supplementary Table S5). Thus, survival supervision was not required to recover prognostic information from the transcriptome. Increasing NMF rank, however, distributed this information across a rank-sensitive set of overlapping factors: at $k = 7$, multiple factors partially captured the same established PDAC signatures, yielding a more fragmented biological representation despite strong performance of the combined predictor (Supplementary Fig. S8).

As a secondary, clinical visualization, a risk cutpoint selected entirely within the training data separated survival across the external validation cohorts (cohort-stratified HR = 1.79, 95% CI 1.41–2.28, dataset-stratified Cox model, $P < 0.001$; Fig. 3b); per-cohort Kaplan–Meier analyses are provided in the Supplementary Information. Prediction alone therefore did not distinguish DeSurv from higher-rank NMF, and, as we show next, comparable prediction is also achieved by conventional survival-supervised models. We therefore asked whether DeSurv instead provided a different biological representation of the prognostic signal, beyond that available from established classifiers and other survival-supervised approaches.

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$), across five validation datasets spanning four independent patient cohorts (combined $n = 570$ unique patients, 388 deaths). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset. Both methods retain significant prognostic value after adjustment.

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.42 (1.26–1.59), < 0.001	1.28 (1.11–1.47), < 0.001
Std NMF ($k = 7$, BO $\alpha = 0$)	1.44 (1.30–1.59), < 0.001	1.40 (1.22–1.62), < 0.001

The survival-aligned structure is reproducible and complementary to established PDAC classifiers

We next asked whether the dominant D1 direction simply recapitulated established PDAC subtype classifications. Across the pooled validation cohorts ($n = 570$, 388 events), binary PurIST and DeCAF classifications together explained 34% of the variance in the continuous D1 score (ordinary R^2 from a linear model with cohort fixed effects, fitted to D1 scores already standardized within cohort), indicating substantial alignment with known tumor-intrinsic and stromal subtypes but incomplete overlap. D1 remained associated with survival after simultaneous adjustment for both classifiers (HR per SD = 0.82, 95% CI 0.73–0.93, $P = 0.0014$); adding D1 to a stratified Cox model containing PurIST and DeCAF improved model fit (partial likelihood-ratio $\chi^2 = 10.1$ on 1 df, $P = 0.0015$) and concordance from 0.592 to 0.622. The DeSurv linear predictor also remained prognostic after adjustment for standard clinical covariates and, where adequately powered, for tumor purity (Supplementary Table S4). Thus, the survival-aligned representation overlaps established tumor and stromal classifications without being reducible to them.

The dominant D1 direction was also not specific to the DeSurv optimizer. Three independent survival-supervised approaches (supervised principal components, Cox partial least squares and sparse Cox) recovered patient scores strongly correlated with D1 in the training cohort ($|r| = 0.81$, 0.83 and 0.85, respectively), whereas the leading unsupervised principal component was nearly orthogonal to D1 ($|r| = 0.10$) and instead tracked higher-variance D2/D3 structure (Supplementary Information, supervised-method recovery). Thus, D1 reflects a reproducible survival-associated expression direction shared across supervised approaches rather than an idiosyncrasy of DeSurv.

Having established D1 as a reproducible survival-associated direction, we asked what DeSurv adds beyond recovering it. Independently tuned supervised principal components and penalized Cox achieved pooled external concordance comparable to DeSurv (C-index 0.60, 0.60 and 0.61, respectively) and captured the dominant D1-associated prognostic direction. Their selected genes, however, were reproducibly enriched for basal-like tumor programs but not proCAF, restCAF, immune or exocrine programs, and their risk scores were weakly correlated with the separately resolved proCAF-like D2 program (penalized Cox, $|r| = 0.06$; supervised PCA, $|r| = 0.14$; Fig. 3c). DeSurv's own linear predictor does not weight D2 either, because elastic-net selection set its coefficient to exactly zero, so D2 enters as a separately measurable program rather than as a component of the fitted risk score. The distinction is therefore not that DeSurv scores stromal signal and the comparators do not, but that DeSurv returns the stromal program as an explicit, transferable object regardless of its weight in the predictor. Although these scores retained modest stromal signal, they did not resolve it as a distinct stromal program (Supplementary Information). Thus, DeSurv's gain is not access to a prognostic direction unavailable to conventional supervised models;

it is the organization of that direction within a fixed, biologically decomposed set of programs whose tumor and stromal components can be measured, interpreted and tested unchanged across cohorts. The comparable survival prediction was achieved largely through a recognizable tumor-intrinsic expression axis rather than a compartmentally resolved representation.

DeSurv recovers prognostic gene programs when variance and prognosis diverge

We used simulations with known latent structure to define when survival supervision alters program recovery. In the primary scenario, the gene program driving survival contributed less transcriptomic variation than outcome-neutral programs ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Materials and Methods, Simulation Studies), so that the strongest sources of expression variation were not the most prognostically relevant. DeSurv and standard NMF ($\alpha = 0$) were tuned using the same cross-validation procedure, isolating the effect of survival supervision during factorization.

Under this setting, DeSurv more accurately recovered the genes defining the true prognostic program, whereas standard NMF largely followed variance-dominant structure (Fig. 4a–b). This distinction is important because the genes assigned to a recovered program, not only its ability to rank patients by risk, determine its subsequent biological interpretation and its potential use in candidate signatures or biomarker development. Across 100 replicates, median test-set C-index was 0.837 for DeSurv versus 0.724 for standard NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$), and DeSurv recovered the true factorization rank ($k = 3$) more frequently (Fig. 4c).

The advantage depended on the relationship between transcriptomic variance and prognosis. It attenuated when prognostic and variance-dominant structure partially overlapped and disappeared under the null, where neither method identified prognostic signal (Supplementary Fig. S4). In the mixed setting, the advantage remained strongest for recovery of the factor-specific marker genes rather than shared background survival genes, supporting a specific effect of supervision on identifying the genes that define the prognostic program. Thus, survival supervision is most useful for biological discovery when clinically relevant programs are not themselves the dominant sources of transcriptomic variation. In this setting, supervision increases the likelihood that the programs carried forward for biological interpretation are also the programs most relevant to outcome, reducing reliance on post hoc screening of variance-dominant factors. These simulations assess recovery under structured perturbations but cannot establish biological accuracy directly; the relevant criterion is whether recovered programs retain coherent compartmental and biological structure.

Discussion

Survival-supervised factorization identified a reproducible three-program tumor–stroma prognostic architecture in PDAC. Its dominant survival-aligned direction, D1, linked classical tumor and restCAF-associated stromal programs, while DeSurv separately resolved proCAF-like stromal and basal-like tumor programs. The principal methodological advance is workflow-level: clinical relevance shapes the programs during discovery, after which the same fixed representation can be tested directly across independent cohorts. The trained representation transferred without refitting across five external validation datasets spanning four independent patient cohorts and remained prognostic after adjustment for established tumor-intrinsic and stromal classifiers. Importantly, DeSurv did not outperform higher-rank NMF or conventional supervised models in external discrimination. Its contribution was instead to place a reproducible prognostic direction within a biologically decomposed tumor–stroma architecture rather than returning either a variance-driven factorization or a single supervised risk score.

The distinction was not simply one of model size or predictive accuracy. Higher-rank NMF achieved comparable external discrimination, but increasingly distributed established PDAC signatures across overlapping factors, with no additional factor mapping cleanly onto a distinct established program. Conventional supervised predictors showed the complementary pattern: they recovered the dominant survival-associated direction efficiently, but their selected genes mapped predominantly to basal-like tumor biology and did not separately resolve the stromal programs. Thus, alternative approaches could recover the prognostic information while representing its underlying biology differently. In simulations, where the latent programs were known, this representational distinction could be evaluated directly: DeSurv more accurately recovered the genes defining the true prognostic program.

This addresses a practical limitation of the conventional discovery sequence. When programs are learned without reference to outcome, their clinical relevance and correspondence across datasets must be established retrospectively through factor selection, biological annotation and cross-study matching. DeSurv does not eliminate the need for external validation, but makes that validation a direct test of programs already shaped by clinical relevance and held fixed across studies.

This distinction is consistent with the structure of the DeSurv objective. Survival supervision acts directly on the gene-program matrix W that holds those weights, while the reconstruction objective requires those same programs to continue representing the observed transcriptome. Once learned, the weights are fixed, and a new tumor is scored by projecting its expression profile onto the top-ranked genes of each program, so the same programs can be evaluated across cohorts without re-estimating the factorization. Empirically, supervision largely retained the major variance-driven tumor and stromal structure represented by D3 and D2 while introducing a D1 program that was poorly reconstructable from the matched-rank NMF factors ($R^2 = 0.09$), at a cost of roughly 6 percentage points in whole-matrix reconstruction R^2. These findings are consistent with survival supervision selectively reorganizing rather than replacing the variance-driven representation, and they place a measurable price on that reorganization rather than presenting it as free. Prior survival-representation approaches, including survival-supervised matrix factorization[28,29] and deep latent-variable models[30,31], have also integrated outcome information with learned representations, but generally do not preserve the same directly transferable gene-program decomposition.

Biologically, D1 represents the dominant survival-aligned program within the broader tumor–stroma architecture rather than the architecture itself. It links classical tumor identity with restCAF-like stromal biology and was recovered de novo without predefined subtype labels. Its classical identity was supported orthogonally by GATA6 RNA in situ hybridization, and its survival association persisted after adjustment for PurIST and DeCAF. Moreover, independent survival-supervised approaches recovered the same D1-associated direction, arguing that D1 reflects reproducible outcome-associated biology rather than an idiosyncrasy of the DeSurv optimizer. The additional D2 and D3 programs provide separately measurable stromal and tumor context around this direction, rather than necessarily representing independent components of the fitted risk score. This organization is consistent with growing evidence that PDAC outcome reflects interactions between malignant lineage state and heterogeneous fibroblast and stromal programs[32–34].

Rank selection represents an additional source of uncertainty in unsupervised decomposition. In the PDAC training data, commonly used NMF criteria gave conflicting rank choices, and increasing rank progressively redistributed established biological signatures across additional factors. Together with the non-uniqueness of reconstruction-based factorization, this rank dependence may contribute to variability among unsupervised molecular taxonomies across studies. Survival supervision does not remove that non-uniqueness, and single DeSurv runs from different random starts recover appreciably different gene sets. What makes the reported decomposition a stable object is the consensus step, and independently seeded replicates of the full procedure recover it closely (Supplementary Information, Stability of the recovered gene programs). Survival-supervised selection therefore does not eliminate factorization uncertainty, but it provides an outcome-relevant criterion for choosing among representations when the scientific objective is specifically to identify clinically relevant programs.

Our simulations clarify when this approach is most useful. When the genes defining the prognostic program contributed relatively little to overall transcriptomic variation, survival supervision improved their recovery; this advantage diminished as prognostic and variance-dominant structure overlapped and disappeared when no survival signal was present. Correct program recovery matters beyond discrimination because the genes assigned to a program determine downstream pathway interpretation, candidate signatures and potential biomarker development. DeSurv is therefore not intended to replace unsupervised decomposition in all settings, but rather to complement it when the scientific goal is to discover biologically interpretable programs aligned with a clinically annotated outcome.

DeSurv is not intended to replace established molecular classifiers or reference-based deconvolution. Its prognostic representation remained informative after adjustment for PurIST and DeCAF (Table 1), indicating complementarity rather than replacement of these compartment-specific classifiers. DeSurv is also distinct from reference-based methods such as CIBERSORTx[35] and BayesPrism[36], which estimate cellular composition using external cell-state references. Instead, DeSurv learns outcome-aligned latent gene programs directly from bulk expression and could in principle be combined with reference-based cell-fraction estimates to refine their cellular attribution.

Several limitations define the scope of these findings. DeSurv currently uses a proportional-hazards survival objective and was evaluated here only in PDAC. Extensions to non-proportional or nonlinear survival relationships are therefore natural methodological directions. More importantly, although the simulations establish conditions under which survival supervision improves recovery of prognostic programs, whether the same representational advantage generalizes across cancer types remains unknown and will require multi-cancer benchmarking against both unsupervised decompositions and strong penalized-regression baselines, which remain difficult to outperform consistently for out-of-sample prognosis[37].

The programs were learned and primarily validated in non-metastatic, treatment-naive disease. The fixed programs also retained clinically relevant associations when projected without retraining into independently treated PDAC cohorts, including metastatic and locally advanced disease (Supplementary Fig. S9), supporting transportability beyond the treatment-naive setting in which they were learned. Because treatment, stage and cohort were not independently varied, these analyses establish prognostic and biological transportability rather than treatment-predictive effects. Arm-separated analyses were underpowered for between-arm contrasts and revealed no reproducible treatment-specific pattern of response or survival, whereas the prognostic association prevailed in the better-powered pooled and arm-stratified analyses; whether DeSurv programs predict differential treatment benefit therefore remains unresolved and will require adequately powered, prespecified treatment-by-program interaction analyses.

The tumor–stroma architecture described here is a bulk-tissue phenomenon. It reflects cross-patient covariation of tumor- and stroma-associated transcriptional programs, not cellular co-expression or spatial adjacency within individual tumors. The biological identity of the survival-aligned

D1 program is nevertheless supported by its persistence after PurIST, DeCAF and tumor-purity adjustment, its recovery by independent survival-supervised approaches, and its correlation with GATA6 RNA in situ hybridization. Mapping these bulk programs onto specific cellular states is not a matter of projecting the same gene weights into single-cell or spatial data. Most of the genes defining each program go undetected in any individual cell, so per-cell projection scores partly reflect detection depth rather than program activity, and multicellular spatial spots mix the compartments the programs are meant to separate. Establishing the cellular correlates of these programs will therefore require clinically annotated single-cell and spatial cohorts together with scoring methods designed for sparse measurement.

More broadly, unsupervised transcriptomic discovery is optimized to recover what varies most, whereas clinically supervised prediction is optimized to recover what predicts outcome; these need not be the same biological structure. In bulk tumors, this mismatch may be compounded by both non-uniqueness of reconstruction-based factorization and ambiguity in selecting factorization rank, potentially contributing to variability among unsupervised molecular taxonomies[2,3,15]. DeSurv addresses this gap by incorporating survival into program discovery while retaining an explicitly decomposed gene-program representation. In PDAC, this revealed a reproducible tumor–stroma prognostic architecture that could be transferred across cohorts, biologically validated and interrogated beyond a risk score alone. DeSurv therefore provides a direct route from clinically annotated bulk transcriptomes to outcome-aligned, biologically interpretable gene programs. By making clinical relevance part of discovery and cross-study transportability a test of the same frozen gene programs, it complements unsupervised deconvolution and conventional supervised prediction without depending on superior predictive accuracy.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p\times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k\times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the event indicators ($\delta_i = 1$ for an observed event, $\delta_i = 0$ for censoring), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k\times n}$: for sample i with projected score $z_i = W^\top x_i$, the hazard is $h(t \mid x_i) = h_0(t)\exp(z_i^\top \beta)$, where h_0 is the baseline hazard. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same gene weights score new cohorts without refitting, the property we exploit for external validation.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1 - \alpha)\, \mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\, \mathcal{L}_{\mathrm{Cox}}(W, \beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. A key design choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[12,38]. This construction makes clinical outcome part of program estimation rather than a post hoc filter applied to an unsupervised decomposition. When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF, and β is fit post hoc by penalized Coxnet on the projected scores $Z = W^\top X$, used only for scoring and cross-validation without affecting the factorization. The Cox component also supports optional adjustment for sample-level covariates (for example, stage or grade); the analyses reported here use factor scores alone.

Optimization alternates updates for H, W, and β (Supplementary Information, Section 2), with backtracking used for proposed W updates. The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Because the implemented algorithm combines normalization, gradient balancing and a Coxnet approximation, we assessed optimization behavior empirically rather than relying on a formal convergence guarantee: across independent random initializations of the reported PDAC model, the fitted objective decreased monotonically to numerically stable solutions (Supplementary Fig. S3). Because objective convergence does not imply that the same programs are recovered each time, we separately assessed program-level stability. Individual restarts are dispersed, whereas the reported consensus procedure recovers the same decomposition across independently seeded replicates (Supplementary Information, Stability of the recovered gene programs). Closed-form update equations are given in Supplementary Information, Sections 1–2, and Bayesian-optimization control parameters in the Bayesian Optimization section.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\mathrm{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)[39,40], which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lies within one standard error of the maximum). All model selection was performed entirely within the training data using cross-validation; no validation cohort data were used during any stage of tuning, and unbiased generalization was assessed only on the external validation cohorts. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (Supplementary Information, Consensus initialization for the final model). Before projection, each learned gene program (column of W) was restricted to its BO-selected number of top genes (denoted $\widetilde{W}$) and column-normalized (ℓ_2-norm scaling) by the same procedure used during training (Supplementary Information, Sections 2 and 5);

the resulting truncated matrix $\widetilde{W}$ was held fixed for all validation cohorts. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor. The ridge penalty on the loadings (H) was held fixed at 0 in all analyses and was not tuned; factor scaling was instead controlled directly by column-normalization of W during optimization.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[13,25]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (Supplementary Fig. S6). Cross-validated concordance varied little across candidate ranks, with no clear optimum anywhere between $k = 2$ and $k = 12$ and a total spread across that range of about two standard errors (Supplementary Fig. S5). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$. This selection is contingent on the gene-selection setting rather than uniquely determined: applying the same rule with all genes retained, instead of the Bayesian-optimization-selected 270 per factor, selects $k = 2$ (Supplementary Fig. S5b). The three-program solution should therefore be read as a parsimonious choice within a flat performance region, with its biological support coming from reference-program correspondence and the GATA6 comparison rather than from a sharp statistical optimum.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^{\top} X_{\text{new}}$, and survival associations were evaluated using Harrell's C-index, hazard ratios, and log-rank statistics; because Harrell's C-index is rank-based and assesses discrimination only, not calibration or time-dependent prediction error, we additionally report a proper score (the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference) as a sensitivity check (Supplementary Information, Proper-scoring sensitivity). Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection requiring no retraining or validation survival information. Related supervised-NMF methods that instead route survival supervision through H require sample-loading re-estimation on each new cohort[28,29]. Validation survival outcomes were therefore used only for downstream performance evaluation, not for model training, scoring, or hyperparameter tuning. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in Supplementary Information, Part I.

To characterize the biological identity of the learned factors, we quantified how strongly each factor's gene loadings aligned with independently defined PDAC tumor and microenvironmental gene programs. For each factor, we correlated its gene loadings with binary membership indicators for each reference program (Spearman correlation over the top 50 genes per factor), assessing significance with Benjamini–Hochberg control of the false discovery rate within each factor. The same analysis was applied to DeSurv and to standard NMF fit at the same rank, and reference programs whose Spearman correlation exceeded 0.2 in at least one factor are shown in the main figure (full details in Supplementary Information).

For orthogonal validation of D1 program identity, we evaluated the association between projected D1 scores and GATA6 RNA in situ hybridization measurements in the COMPASS cohort among the 33 tumors with both measurements, using Spearman correlation. Because *GATA6* is itself one of the 805 genes contributing to the D1 projection, part of this correlation could reflect the score being built from the assayed gene. We therefore repeated the comparison with *GATA6* removed from the projection, and the correlation was essentially unchanged (Spearman $\rho = 0.67$ versus 0.68), indicating the association is not an artifact of self-correlation.

Supervised comparator analyses

To distinguish prognostic discrimination from biological representation, we independently fit supervised principal components (superpc; cross-validated screening threshold and component number) and lasso-penalized Cox regression (glmnet; λ_{min}) using the same 1,970-gene training universe as DeSurv. Each method performed feature selection and tuning using the training data only, after which the fitted model was frozen and applied to rank-transformed validation cohorts without refitting. The two comparators select features by different mechanisms, and neither matches DeSurv's. The penalized Cox comparator uses a pure lasso penalty (`glmnet` with $\alpha = 1$), whereas supervised principal components applies no penalty at all, instead screening genes by a cross-validated feature-score threshold and extracting a cross-validated number of components. DeSurv's internal Cox step uses an elastic net whose Bayesian-optimization-selected mixing parameter ($\xi = 0.056$) sits close to ridge. Both comparators are therefore far sparser in the genes they retain than DeSurv's 810-gene support, and both received a smaller tuning budget, a single internal cross-validation each against DeSurv's Bayesian-optimization search and consensus initialization, which should be borne in mind when reading their comparable external discrimination. Correspondence between each supervised risk score and the DeSurv programs D1–D3 was quantified in the training cohort using absolute Spearman correlation. Additional analyses of selected-gene enrichment and compartment attribution are described in the Supplementary Information.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates using test-set concordance and gene-program recovery. Gene-program recovery was quantified for each survival-driving factor by identifying the learned factor whose top-ranked genes had the highest precision against that factor's true program (the best-matching learned factor); recovery was then the proportion of that factor's top-n_{top} genes (ranked by factor-specificity score, with n_{top} the Bayesian-optimization-selected signature size) belonging to the ground-truth prognostic gene set, averaged across survival-driving factors. In the primary scenario the target comprised the 150 factor-specific marker genes driving survival ($\beta = (2, 0, 0)$); in the mixed scenario, marker-only and full survival-gene definitions of the target were evaluated separately. We considered three simulation scenarios: a prognostic scenario in which prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available and controlled-access RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Eligible samples were primary PDAC tumors with a measured bulk expression profile and valid overall-survival annotation, subject to cohort-specific quality-control and exclusion criteria detailed in Supplementary Information, Section 6; accrual and follow-up periods are as defined in each cohort's original publication. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. No missing-data imputation was performed: all analyses used complete cases, and analyses adjusted for the PurIST and DeCAF classifiers were restricted to samples with available classifier calls. Of the seven PDAC expression datasets we considered, two were used for training (TCGA and CPTAC) and five for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo); because the two PACA-AU platforms profile one patient cohort, these five validation datasets span four independent patient cohorts. To harmonize differences in scale across cohorts, filtered gene expression data were within-sample rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility); ranking removes platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. More details about the datasets can be found in Supplementary Information, Section 6.

Per-cohort participant flow is summarized in Supplementary Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, missing tumor grade, or quality-control failure, yielding 273 analytic training samples (TCGA-PAAD 144, CPTAC-3 129; 139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation profiles across five platform datasets: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181). PACA-AU included partially overlapping array and RNA-seq datasets (46 shared patients). Platform-specific analyses retained each dataset separately; for analyses combining validation datasets, duplicate patient accessions were represented once, preferentially retaining the RNA-seq profile and excluding the corresponding array profile, consistent with the preprocessing rule used in the previously published DeCAF analysis[5]. This yielded a combined validation population of 570 unique patients (388 deaths) spanning four independent patient cohorts.

Use of large language models

During preparation of this manuscript, the authors used a large language model (Claude; Anthropic) to assist with language editing and with code for generating and formatting the display items. All such output was directed, reviewed, and verified by the authors, who take full responsibility for the work. The model was not used to design the study, develop the DeSurv methodology, or interpret the findings, and it does not meet the criteria for authorship.

Data Availability

The training and validation datasets used in this study are publicly available or accessible through controlled-access repositories, and are de-identified; no Institutional Review Board (IRB) approval was required for their analysis. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[41] and CPTAC-3[42]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[43]; Puleo, E-MTAB-6134[44]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[45]), and the International Cancer Genome Consortium (ICGC): the PACA-AU array cohort is publicly available from the Gene Expression Omnibus (GSE36924), and the PACA-AU RNA-seq cohort from the European Genome–phenome Archive (EGA) study EGAS00001000154 under controlled access[46]. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5]; cohort details and sample sizes after quality control are summarized in Supplementary Information, Section 6. The treated bulk-tissue cohorts used in the exploratory translational analysis (Rash-Accept and the Linehan FOLFIRINOX$\pm$CCR2-inhibitor series), and the O'Kane/COMPASS treated metastatic PDAC cohort used both for the D1 prognostic-transportability survival analysis (Supplementary Fig. S9A) and, in the subset of 33 tumors with matched staining, for the GATA6 RNA in situ hybridization comparison (main-text Fig. 2e), are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and Supplementary Information are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified. Code used to generate the COMPASS D1–GATA6 comparison (main-text Fig. 2e) is also provided in the reproducibility repository; reproduction of the patient-level analysis requires authorized access to the COMPASS data.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), `NMF`[38], `glmnet`[47], `survival`, `ggplot2`, and `cowplot`; Bayesian optimization for hyper-parameter selection was implemented within DeSurv. The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and the exact session-info file are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

Acknowledgements

This work was supported by the National Cancer Institute grants U01 CA274298, P50 CA257911, T32 CA106209, and R01 CA288145, and by the Department of Defense Peer Reviewed Cancer Research Program (HT9425241103100121). We thank the patients and research teams who contributed samples and clinical data to the public cohorts used here (TCGA, CPTAC, Dijk, Moffitt, Puleo, PACA-AU).

Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide. Sex was recorded in every cohort and was included as a covariate in the clinical-adjustment analyses (Supplementary Table S4), where the prognostic association of the DeSurv linear predictor was unchanged. We did not conduct sex-stratified analyses, because the study was not powered for subgroup comparisons within individual cohorts, and we did not stratify by race or ethnicity because those annotations were not consistently available across cohorts. Whether the prognostic relevance of these gene programs differs by sex, race, or ethnicity remains an important question for adequately powered, more comprehensively annotated cohorts.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and included as fixed covariate annotations, and the adjustment analyses using these calls are objective and reproducible from the deposited code. They are not components of DeSurv. The other authors declare no competing interests.

References

1. Siegel, R. L., Kratzer, T. B., Wagle, N. S., Sung, H. & Jemal, A. Cancer statistics, 2026. *CA: A Cancer Journal for Clinicians* https://doi.org/10.3322/caac.70043 (2026) doi:10.3322/caac.70043.
2. Collisson, E. A. *et al.* Subtypes of pancreatic ductal adenocarcinoma and their differing responses to therapy. *Nature Medicine* **17**, 500–503 (2011).
3. Moffitt, R. A. *et al.* Virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma. *Nature Genetics* **47**, 1168–1178 (2015).
4. Rashid, N. U. *et al.* Purity independent subtyping of tumors (PurIST), a clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. *Clinical Cancer Research* **26**, 82–92 (2020).
5. Peng, X. L. *et al.* DeCAF defines clinical fibroblast subtypes and multidimensional tumor-stroma crosstalk shaping prognosis and immunotherapy response. *Cell Reports Medicine* **7**, 102611 (2026).
6. Moncada, R. *et al.* Integrating microarray-based spatial transcriptomics and single-cell RNA-seq reveals tissue architecture in pancreatic ductal adenocarcinomas. *Nature Biotechnology* **38**, 333–342 (2020).
7. Hwang, W. L. *et al.* Single-nucleus and spatial transcriptome profiling of pancreatic cancer identifies multicellular dynamics associated with neoadjuvant treatment. *Nature Genetics* **54**, 1178–1191 (2022).
8. Werba, G. *et al.* Single-cell RNA sequencing reveals the effects of chemotherapy on human pancreatic adenocarcinoma and its tumor microenvironment. *Nature Communications* **14**, 797 (2023).
9. Bair, E. & Tibshirani, R. Semi-supervised methods to predict patient survival from gene expression data. *PLoS Biology* **2**, e108 (2004).
10. Aung, K. L. *et al.* Genomics-driven precision medicine for advanced pancreatic cancer: Early results from the COMPASS trial. *Clinical Cancer Research* **24**, 1344–1354 (2018).
11. Nguyen, H., Nguyen, H., Tran, D., Draghici, S. & Nguyen, T. Fourteen years of cellular deconvolution: Methodology, applications, technical evaluation and outstanding challenges. *Nucleic Acids Research* **52**, 4761–4783 (2024).
12. Lee, D. D. & Seung, H. S. Learning the parts of objects by non-negative matrix factorization. *Nature* **401**, 788–791 (1999).
13. Brunet, J.-P., Tamayo, P., Golub, T. R. & Mesirov, J. P. Metagenes and molecular pattern discovery using matrix factorization. *Proceedings of the National Academy of Sciences USA* **101**, 4164–4169 (2004).
14. Peng, X. L., Moffitt, R. A., Torphy, R. J., Volmar, K. E. & Yeh, J. J. De novo compartment deconvolution and weight estimation of tumor samples using DECODER. *Nature Communications* **10**, 4729 (2019).
15. Bailey, P. *et al.* Genomic analyses identify molecular subtypes of pancreatic cancer. *Nature* **531**, 47–52 (2016).
16. Cook, R. D. Fisher lecture: Dimension reduction in regression. *Statistical Science* **22**, 1–26 (2007).

17. Torre-Healy, L. A. *et al.* Open-source curation of a pancreatic ductal adenocarcinoma gene expression analysis platform (pdacR) supports a two-subtype model of clinical relevance. *Communications Biology* **6**, 163 (2023).

18. Aran, D., Sirota, M. & Butte, A. J. Systematic pan-cancer analysis of tumour purity. *Nature Communications* **6**, 9971 (2015).

19. Donoho, D. & Stodden, V. When does non-negative matrix factorization give a correct decomposition into parts? in *Advances in neural information processing systems 16 (NIPS 2003)* (eds. Thrun, S., Saul, L. K. & Schölkopf, B.) 1141–1148 (MIT Press, 2004).

20. Laurberg, H., Christensen, M. G., Plumbley, M. D., Hansen, L. K. & Jensen, S. H. Theorems on positive data: On the uniqueness of NMF. *Computational Intelligence and Neuroscience* **2008**, 764206 (2008).

21. Gillis, N. The why and how of nonnegative matrix factorization. in *Regularization, optimization, kernels, and support vector machines* (eds. Suykens, J. A. K., Signoretto, M. & Argyriou, A.) 257–291 (Chapman; Hall/CRC, 2014).

22. Lautizi, M., Baumbach, J., Weichert, W., Steiger, K. & List, M. A systematic assessment of pancreatic cancer molecular subtype classification across cohorts. *NAR Cancer* **4**, zcac030 (2022).

23. Raphael, B. J. *et al.* Integrated genomic characterization of pancreatic ductal adenocarcinoma. *Cancer Cell* **32**, 185–203 (2017).

24. Cao, L. *et al.* Proteogenomic characterization of pancreatic ductal adenocarcinoma. *Cell* **184**, 5031–5052 (2021).

25. Frigyesi, A. & Höglund, M. Non-negative matrix factorization for the analysis of complex gene expression data: Identification of clinically relevant tumor subtypes. *Cancer Informatics* **6**, 275–292 (2008).

26. Leary, J. R. *et al.* Sub-cluster identification through semi-supervised optimization of rare-cell silhouettes (SCISSORS) in single-cell RNA-sequencing. *Bioinformatics* **39**, btad449 (2023).

27. Maurer, C. *et al.* Experimental microdissection enables functional harmonisation of pancreatic cancer subtypes. *Gut* **68**, 1034–1043 (2019).

28. Huang, Z., Salama, P., Shao, W., Zhang, J. & Huang, K. Low-rank reorganization via proportional hazards non-negative matrix factorization unveils survival associated gene clusters. (2020).

29. Le Goff, V. *et al.* SurvNMF: Non-negative matrix factorization supervised for survival data analysis. (2025).

30. Zaccaria, G. M., Altini, N., Mezzolla, G., *et al.* SurvIAE: Survival prediction with interpretable autoencoders from diffuse large B-cell lymphoma gene expression data. *Computer Methods and Programs in Biomedicine* **244**, 107966 (2024).

31. Meng, X., Wang, X., Zhang, X., *et al.* A novel attention-mechanism based Cox survival model by exploiting pan-cancer empirical genomic information. *Cells* **11**, 1421 (2022).

32. Ogawa, Y., Masugi, Y., Abe, T., *et al.* Three distinct stroma types in human pancreatic cancer identified by image analysis of fibroblast subpopulations and collagen. *Clinical Cancer Research* **27**, 107–119 (2021).

33. Neuzillet, C., Nicolle, R., Raffenne, J., *et al.* Periostin- and podoplanin-positive cancer-associated fibroblast subtypes cooperate to shape the inflamed tumor microenvironment in aggressive pancreatic adenocarcinoma. *The Journal of Pathology* **258**, 408–425 (2022).

34. Lee, J. J., Kearney, J. F., Trembath, H. E., *et al.* Tumor-intrinsic and cancer-associated fibroblast subtypes independently predict outcomes in pancreatic cancer. *Annals of Surgery* **280**, 659–666 (2024).

35. Newman, A. M., Steen, C. B., Liu, C. L., Gentles, A. J., *et al.* Determining cell type abundance and expression from bulk tissues with digital cytometry. *Nature Biotechnology* **37**, 773–782 (2019).

36. Chu, T., Wang, Z., Pe'er, D. & Danko, C. G. Cell type and gene expression deconvolution with BayesPrism enables Bayesian integrative analysis across bulk and single-cell RNA sequencing in oncology. *Nature Cancer* **3**, 505–517 (2022).

37. Smith, A. M., Walsh, J. R., Long, J., *et al.* Standard machine learning approaches outperform deep representation learning on phenotype prediction from transcriptomics data. *BMC Bioinformatics* **21**, 119 (2020).

38. Gaujoux, R. & Seoighe, C. A flexible r package for nonnegative matrix factorization. *BMC Bioinformatics* **11**, 367 (2010).

39. Snoek, J., Larochelle, H. & Adams, R. P. Practical Bayesian optimization of machine learning algorithms. in *Advances in neural information processing systems 25 (NeurIPS 2012)* (2012).

40. Shahriari, B., Swersky, K., Wang, Z., Adams, R. P. & Freitas, N. de. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE* **104**, 148–175 (2016).

41. The Cancer Genome Atlas Research Network. Data from 'integrated genomic characterization of pancreatic ductal adenocarcinoma' (TCGA-PAAD). (2017).

42. National Cancer Institute Clinical Proteomic Tumor Analysis Consortium. Data from CPTAC-3 pancreatic ductal adenocarcinoma cohort. (2019).

43. Dijk, F. *et al.* Data from 'unsupervised class discovery in pancreatic ductal adenocarcinoma'. (2020).

44. Puleo, F. *et al.* Data from 'stratification of pancreatic ductal adenocarcinomas based on tumor and microenvironment features'. (2018).

45. Moffitt, R. A. *et al.* Data from 'virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma'. (2015).

46. Australian Pancreatic Cancer Genome Initiative & International Cancer Genome Consortium. Pancreatic cancer (PACA-AU) cohort: Array and RNA-sequencing data. (2016).

47. Simon, N., Friedman, J. H., Hastie, T. & Tibshirani, R. Regularization paths for cox's proportional hazards model via coordinate descent. *Journal of Statistical Software* **39**, 1–13 (2011).

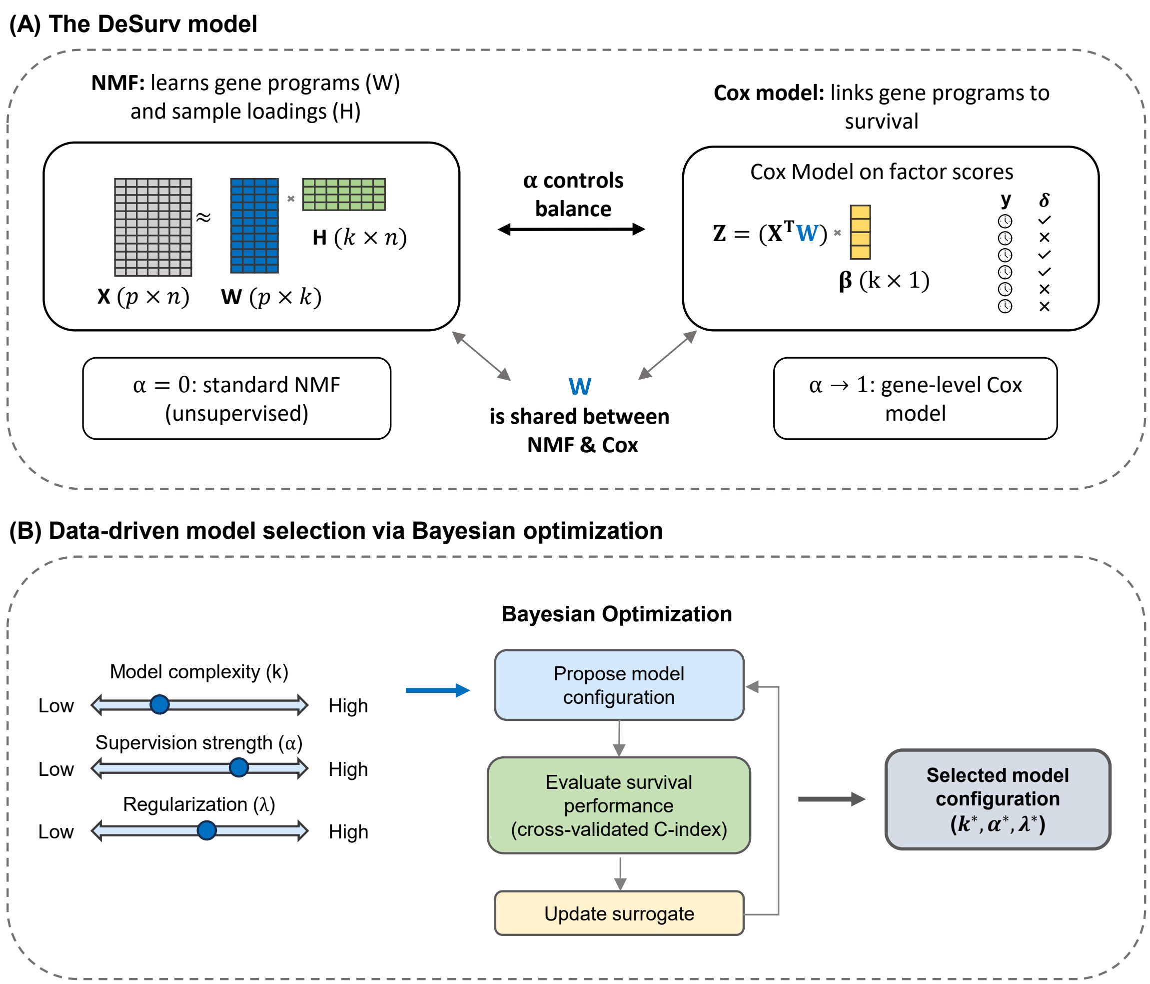

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index. Survival supervision acts on the gene-program matrix W during discovery, rather than being applied only after an unsupervised factorization has been completed; following training-only model selection, the resulting gene weights are fixed for direct evaluation in external cohorts.

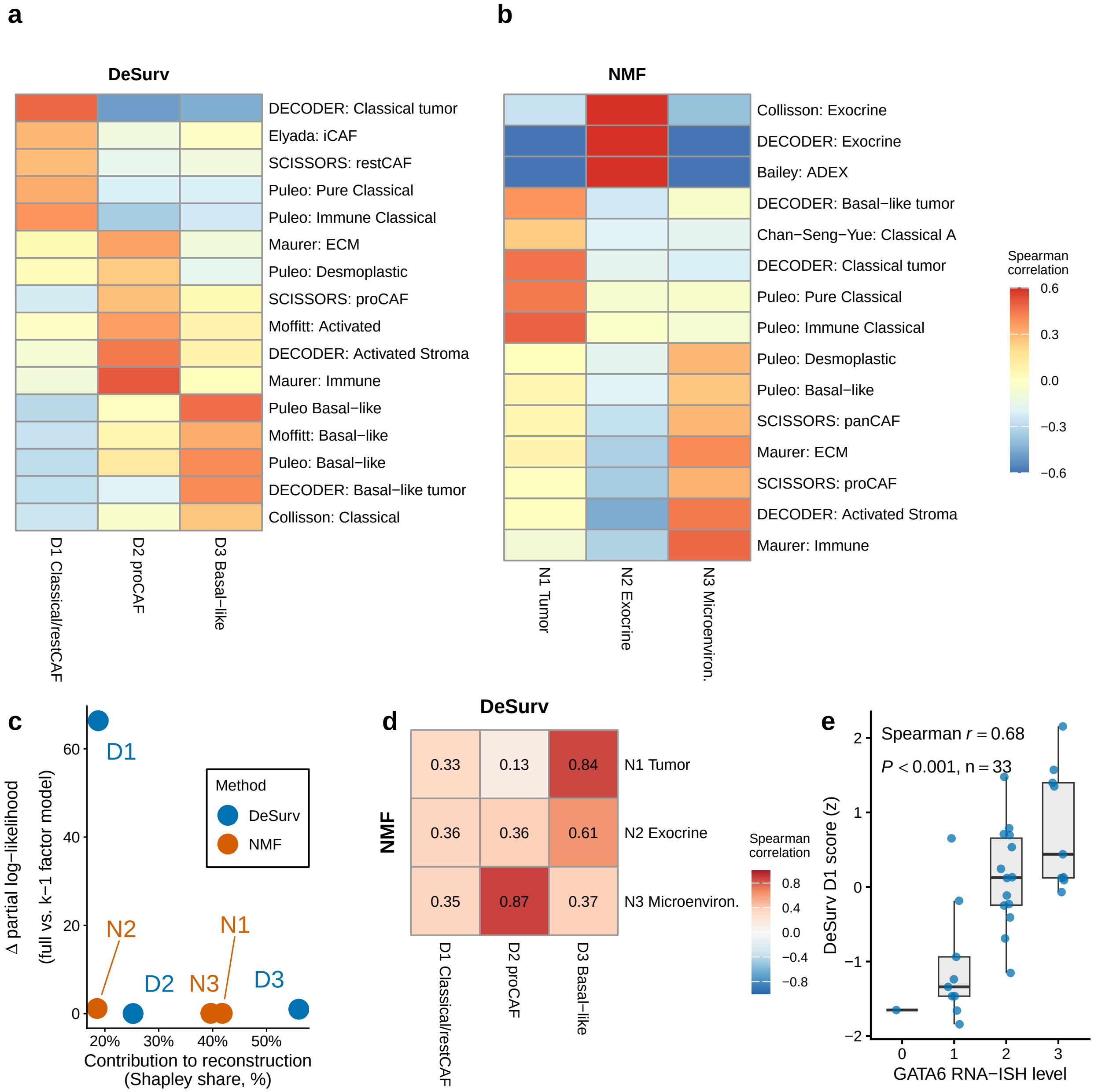

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (a,b) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (a) and standard NMF (b) at $k = 3$; reference programs with $r > 0.2$ for at least one factor are shown. (c) Per-factor reconstruction contribution (normalized Shapley share, summing to 100%) versus survival contribution (Δ Cox partial log-likelihood on factor removal). (d) Pairwise correspondence between NMF and DeSurv factors (Spearman correlation of W-matrix gene loadings over all genes). (e) DeSurv D1 score versus GATA6 RNA in situ hybridization level in the COMPASS cohort (Spearman correlation; $n = 33$).

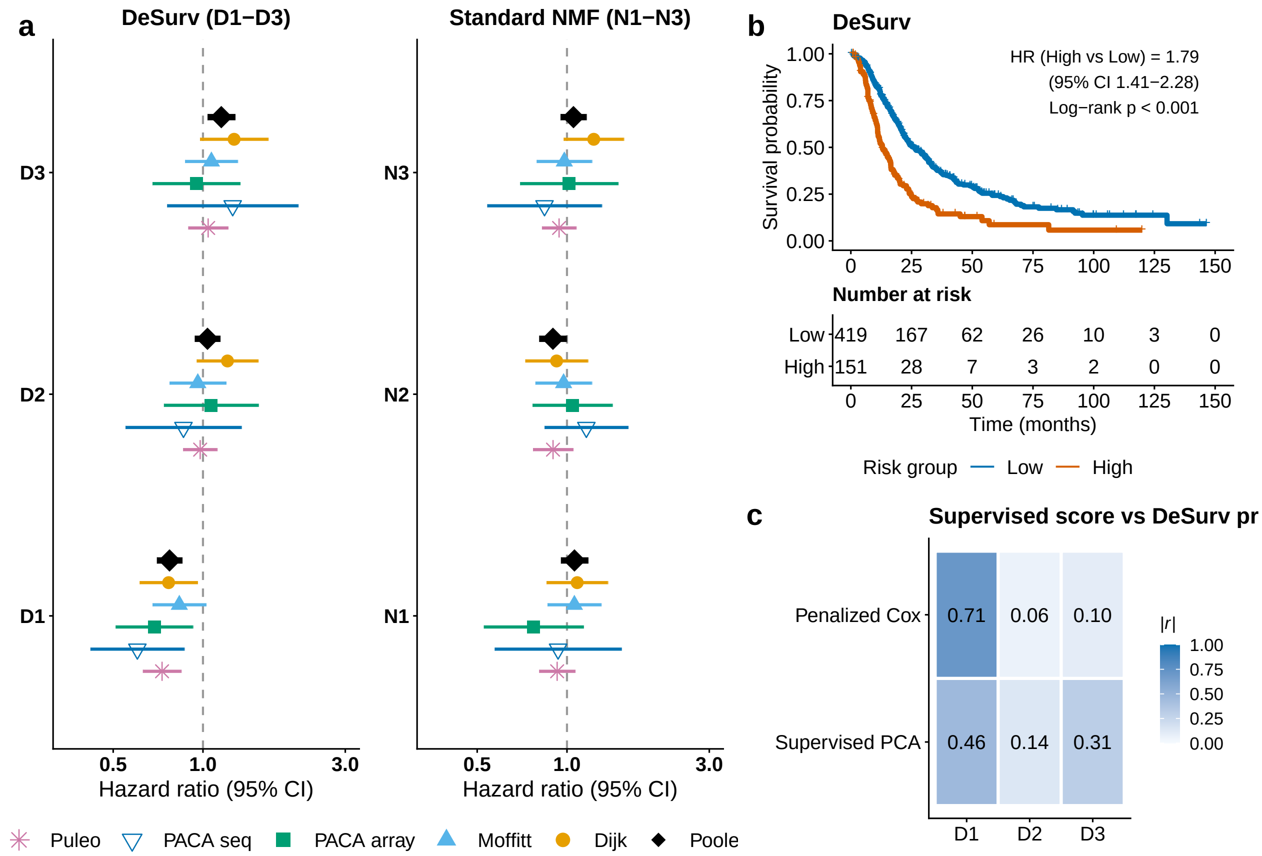

Figure 3: Survival-aligned programs generalize across independent PDAC patient cohorts (five external validation datasets: Dijk, Moffitt GEO array, PACA-AU array, PACA-AU RNA-seq and Puleo; combined $n = 570$ unique patients, 388 deaths, with PACA-AU patients profiled on both platforms represented once via RNA-seq). (a) Forest plot of per-dataset univariate hazard ratios (95% CIs) for each factor; the platform-specific PACA-AU array and RNA-seq estimates are shown separately, and the black diamonds denote the combined estimate over unique patients (per-factor cohort-stratified Cox on the de-duplicated patient-level data; PACA-AU array duplicates excluded). (b) Kaplan–Meier curves for the combined validation samples stratified by the training-derived multivariate DeSurv linear predictor, dichotomized at a cross-validated optimal cutpoint; the pooled curves are shown for visualization, while the reported hazard ratio and log-rank P value use cohort-stratified inference. (c) Correspondence ($|$Spearman $r|$) of tuned supervised risk scores (penalized Cox and supervised PCA) with the three DeSurv programs (D1–D3); supervised scores concentrate on D1 and are near-orthogonal to the D2 stromal program. No validation-cohort survival outcomes were used in training, scoring or hyperparameter selection.

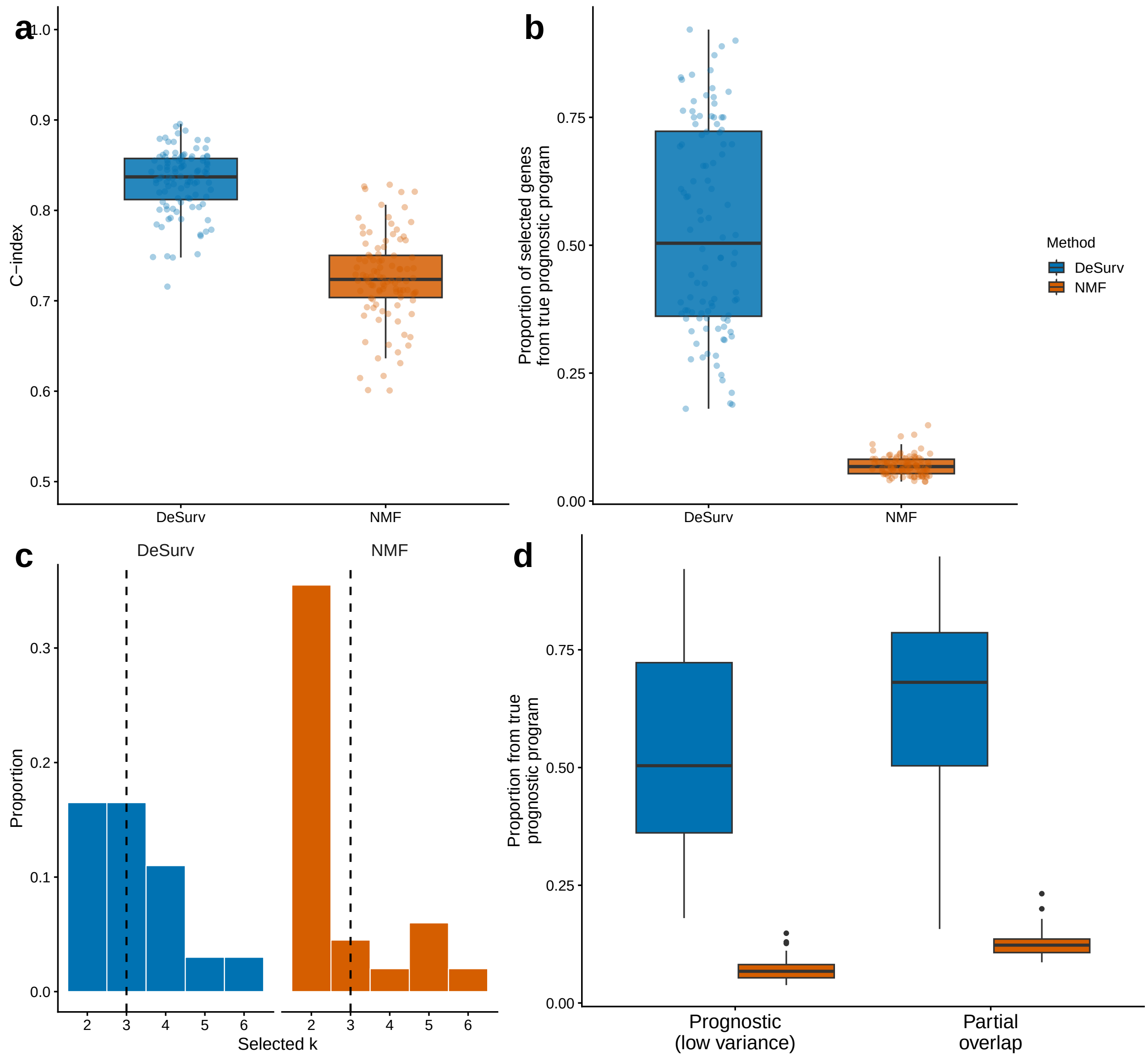

Figure 4: Survival supervision improves recovery of prognostic gene programs when variance and prognosis diverge. (a) Test-set concordance across 100 simulation replicates comparing DeSurv with standard NMF ($\alpha = 0$; $p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; paired Wilcoxon $P < 0.001$). (b) Recovery of genes defining the true prognostic program, quantified as the proportion of selected genes in the best-matching learned program that belonged to the ground-truth prognostic gene set. In (a) and (b), boxes show median and interquartile range; whiskers extend to 1.5×IQR; points are individual replicates. (c) Distribution of selected ranks across replicates for DeSurv versus standard NMF. (d) The same gene-program recovery metric as in (b), computed for two signal regimes: a prognostic program that explains low transcriptomic variance (separated from the dominant variance) versus one that partially overlaps the dominant variance. DeSurv recovers the prognostic program in both regimes, whereas standard NMF, which follows variance, does not; the null scenario (no prognostic program) is shown in the Supplementary Information.